

Global Warming: Impact & Solutions

1. Introduction

Global warming refers to the long-term rise in the average temperature of Earth's climate system. It is a major aspect of climate change and has been demonstrated by direct temperature measurements and by measurements of various effects of the warming. The term commonly refers to the mainly human-caused observed warming since pre-industrial times and its projected continuation.

The Greenhouse Effect: Natural atmospheric gases like CO_2 and CH_4 trap solar radiation. While essential for life, human-induced excess of these gases creates an insulating blanket that overheats the planet.

2. Major Causes

A. Greenhouse Gas Emissions

The combustion of fossil fuels (coal, oil, and natural gas) for electricity, heat, and transportation is the primary source of CO_2 emissions.

B. Deforestation

Trees act as "carbon sinks" by absorbing CO_2 . Mass clearing of forests for agriculture or urban development reduces the planet's capacity to remove carbon from the atmosphere.

C. Industrial & Agricultural Activity

Modern farming practices release large amounts of Methane (CH_4) and Nitrous Oxide (N_2O), both of which have a higher heat-trapping potential than carbon dioxide.

3. Consequences and Impacts

- **Rising Sea Levels:** The melting of polar ice caps and thermal expansion of seawater threaten to submerge coastal cities.

- **Extreme Weather:** Increased global temperatures lead to more intense heatwaves, severe droughts, and powerful storms.
- **Biodiversity Loss:** Rapidly changing habitats make it difficult for many species to survive, leading to increased extinction rates.

4. Quick Reference Summary

Key Metric	Current Status / Target
Current Warming	~1.1°C above pre-industrial levels
Critical Threshold	1.5°C (Paris Agreement goal)
Primary Contributor	Carbon Dioxide (CO ₂)
Main Source	Energy Production & Transportation

5. Mitigation and Solutions

Limiting global warming requires a coordinated global effort across several sectors:

- **Transition to Renewables:** Replacing fossil fuels with wind, solar, and hydroelectric power.
- **Energy Efficiency:** Implementing smarter building designs and efficient industrial processes.
- **Policy and Regulation:** International treaties like the Paris Agreement and domestic carbon taxes to incentivize lower emissions.